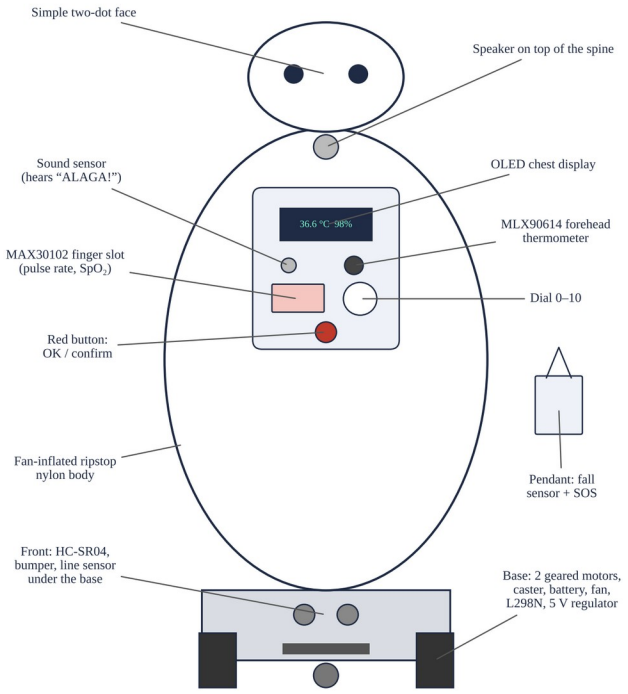
ALAGA

Build Guide: Materials and Step-by-Step Construction

A low-cost wheeled soft companion robot for older adults home alone (RIM-T)



Front view, not to scale. About 1.35 m tall when inflated.

ALAGA when finished: about 1.35 m tall, white, soft, and on wheels.

What it does. While the family is at work, ALAGA waits flat at its dock. At set times it inflates, drives along a tape line to Lola's or Lolo's chair, and reminds them to take medicine (they press the red button after), drink water, or answer a check-in. Twice a day it checks temperature, pulse rate, oxygen (SpO_2), and pain. It plays music and prayers when called. Every result goes to the family's Telegram group. A pendant worn on a cord detects falls and has an SOS button; a fall or SOS alerts the family within seconds.

Time needed. About 5 weekends. **Estimated cost:** about PHP 6,600 for the robot and pendant (Section 1). Home Wi-Fi is needed for the family messages.

Important. ALAGA is a school research prototype, not a medical device and not a replacement for a caregiver. It reminds, checks, and alerts; it never diagnoses or gives medicine. Test on people only after the school IRB approves the plan and each participant signs a consent form (ISEF Form 4). Never ask an older adult to simulate a fall.

1. Materials

Prices are estimates from Philippine electronics shops and online stores (Shopee, Lazada, e-Gizmo, Makerlab) in October 2026. Keep every official receipt; the paper's Appendix A uses the same items.

Part and specification	Qty	Est. PHP
Ripstop nylon 70D, white (per meter) (150 cm wide, PU-coated)	3	450
Seam tape, zipper, thread, felt	1 lot	200
12 V DC blower fan with guard	1	250
12 V JGA25-370 geared motor with bracket (12 V, about 130 rpm, with mounting bracket)	2	700
65-mm wheels with couplers (pair); 2-in caster (4-mm or 6-mm couplers to match the motor shaft)	1 set	210
L298N motor driver	1	120
5-channel line sensor; 2 bumper microswitches (TCRT5000 5-way tracking module; lever microswitches)	1 set	220
ESP32 development board	1	380
MLX90614 (GY-906) module	1	550
MAX30102 module	1	180
HC-SR04 ultrasonic sensor	1	70
0.96-in OLED display	1	180
DFPlayer Mini, 8 GB microSD, 3 W speaker	1 set	380
KY-037 sound sensor, button, 10 kΩ dial	1 set	160
12 V 7 Ah SLA battery	1	650
LM2596, D4184 module, fuse, switch, diode	1 set	180
Pendant: ESP32-C3, MPU6050, LiPo, TP4056, buzzer, case (ESP32-C3 SuperMini, GY-521, 400 mAh 3.7 V LiPo, active 3 V buzzer, 12-mm button, small box, cord)	1 set	600
Wires, perfboard, chest box, plywood, PVC, screws (20 × 15 × 7 cm project box; 9-mm plywood; 3/4-in PVC, T, end cap, floor flange; 10-core ribbon cable; 1 kΩ ×2, 2 kΩ ×1; grommet kit)	1 lot	550
Black electrical tape, 19 mm	3 rolls	120
Rigid shell for the usability test	1	300
70% alcohol, wipes, data sheets	1 lot	150
TOTAL (estimated)		**6,600**

Tools and equipment (borrow if possible)

- Sewing machine, fabric scissors, measuring tape, tailor's chalk, pins
- Soldering iron and solder, wire stripper, screwdriver set, multimeter, hot-glue gun
- Drill with bits, jigsaw or hand saw, PVC cutter
- Laptop with Arduino IDE 2; a phone with Telegram; a phone for recording the voice
- 12 V SLA battery charger (1–2 A) and a USB charger for the pendant
- For calibration and tests: clinic forehead thermometer and fingertip pulse oximeter, probe thermometer, pot and black-painted can, two 20-cm gym mats, knee and elbow pads, stopwatch, tape measure

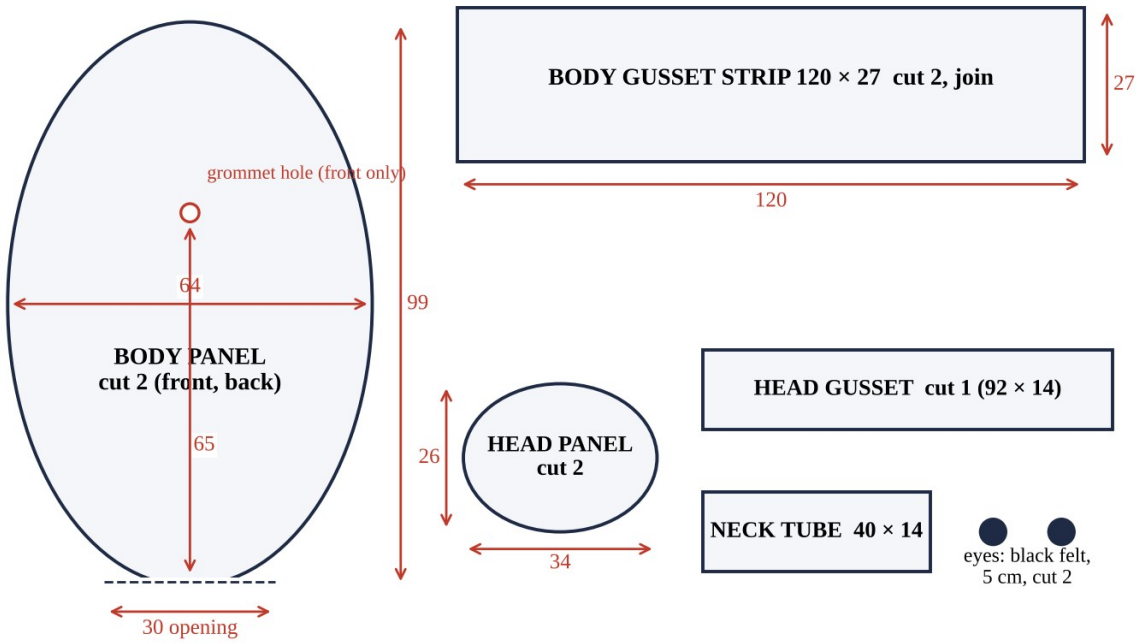
Safety rules while building

- Solder only with an adult present, in a ventilated room.
- Set the LM2596 output to 5.0 V with a multimeter BEFORE connecting anything to it.
- Put the 3 A fuse on the battery + wire. Never short the battery or the LiPo.

- Lift the wheels off the floor the first time you test the motors.
- Keep fingers away from the fan and the wheels.

2. Sew the Inflatable Body

The body is a constant-air inflatable, like an inflatable costume: the fan runs while ALAGA is working and air slowly leaks out. Seams do not need to be airtight, but they must be strong.



All sizes in centimeters and include a 1-cm seam allowance. Fabric: white 70D ripstop nylon, 150 cm wide, 3 m. Long strips are drawn shorter than their real lengths.

Figure A. Cutting pattern (cm). Draw each oval by marking its width and height and joining the points with a smooth curve.

1. Mark and cut: two body panels (64 × 99 cm ovals with a 30-cm flat opening at the bottom), two body gusset strips (120 × 27 cm), two head panels (34 × 26 cm ovals), one head gusset (92 × 14 cm), one neck tube (40 × 14 cm), and two 5-cm black felt eyes.
2. Sew the two gusset strips end to end into one strip about 238 cm long.
3. Front panel: 65 cm above the bottom opening, glue a 10 × 10 cm patch inside and fix the 25-mm grommet through both layers. The chest box is bolted to the spine through this hole.
4. Sew the gusset around the front panel from one side of the bottom opening, over the top, to the other side. Sew the back panel to the other edge of the gusset. Trim extra strip. Turn right side out.
5. Cut a 12-cm neck hole at the top of the gusset. Sew the head (two panels and the head gusset) with a 12-cm hole underneath, and join the two holes with the neck tube so air reaches the head.
6. Glue or sew the two black felt eyes about 6 cm apart.
7. Hem the bottom opening (fold 2 cm). It will be stapled to the top of the base.

3. Build the Wheeled Base and Spine

1. Cut 9-mm plywood: bottom and top 40 × 40 cm; four sides 40 × 12 cm (two of them 38.2 × 12 cm). Glue and screw the sides to the bottom. Keep the top removable (4 screws).
2. Motors: bolt one motor bracket under each rear corner so the wheels stick out about 1 cm past the sides and touch the floor. Fit the wheels with the couplers.
3. Caster: screw the 2-in swivel caster under the front center. Adjust with washers until the base sits level with the line sensor about 1 cm above the floor.
4. Front: screw the 5-channel line sensor under the front edge, sensors down, 1 cm above the floor. Mount the HC-SR04 on the front side panel, pointing forward. Make a bumper from a 30-cm strip of thin plywood on two lever microswitches across the front, so a light push closes a switch.
5. Inside: fix the battery between the wheels (as low as possible), then the fuse holder, switch, LM2596, L298N, and D4184. Drill the switch hole in a side panel.
6. Fan: cut a hole in the top board for the blower outlet, 8 cm from the center, and hot-glue the blower under it, outlet up. Drill air holes in a side panel near the fan intake and fit the finger guard.
7. Spine: screw the PVC floor flange to the center of the top board. Build the spine: 62-cm pipe, T (side outlet facing forward), 25-cm pipe on top. Add a 6-cm pipe and end cap to the T. Run the ribbon cable and the 5 V/GND wires up the pipe and out through a hole in the end cap.
8. Pull the body over the spine so the end cap meets the grommet, and staple the bottom hem around the edge of the top board every 3 cm. Zip-tie the speaker to the top of the spine.

4. Wire the Electronics

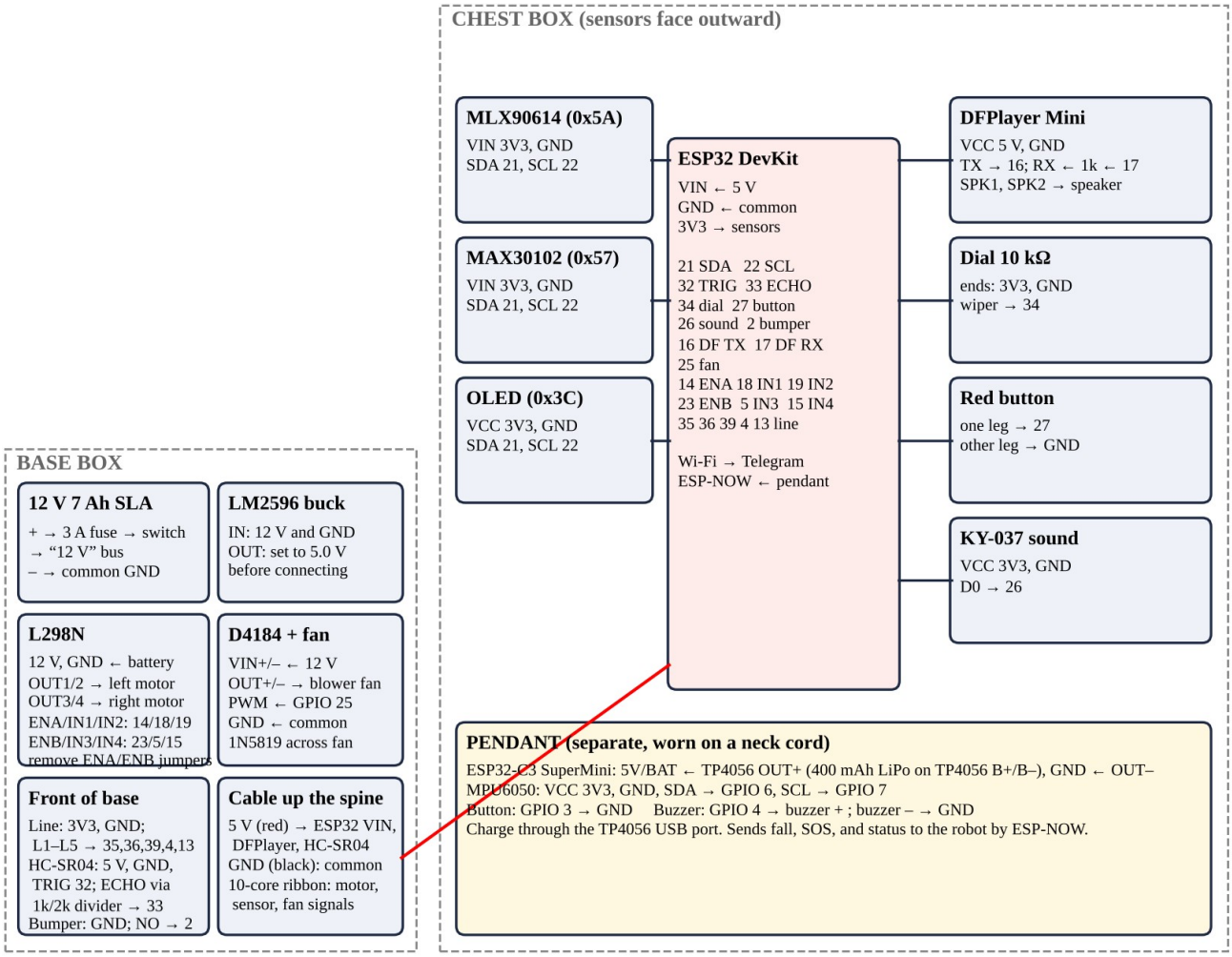


Figure B. Wiring of ALAGA and its pendant. All grounds (GND) are joined together.

1. Power: battery + → 3 A fuse → rocker switch → 12 V bus; battery – → GND bus. Connect the 12 V and GND buses to the LM2596 input, the L298N +12V/GND, and the D4184 VIN.
2. Switch on, turn the LM2596 screw until its output reads 5.0 V, switch off. Then connect 5 V to ESP32 VIN, DFPlayer VCC, and HC-SR04 VCC.
3. Motors: left motor to L298N OUT1/OUT2, right motor to OUT3/OUT4. Remove the ENA and ENB jumpers and wire ENA, IN1, IN2, ENB, IN3, IN4 to GPIO 14, 18, 19, 23, 5, 15.
4. Fan: D4184 OUT+/OUT– to the fan, 1N5819 across the fan (stripe to +), PWM to GPIO 25.
5. Front sensors: line sensor L1 (left) to L5 (right) → GPIO 35, 36, 39, 4, 13, powered from 3V3. HC-SR04 TRIG → 32; ECHO → 1 kΩ → GPIO 33, and GPIO 33 → 2 kΩ → GND. Bumper switches: COM to GND, NO to GPIO 2.
6. Chest box: cut holes for the OLED, MLX90614 (near the top), finger slot, dial, red button, sound sensor, and cable. Solder the ESP32 on perfboard with female headers. Wire the I²C parts

(MLX90614, MAX30102, OLED) to 3V3, GND, SDA 21, SCL 22; the DFPlayer TX → 16 and RX ← 1 kΩ ← 17; dial wiper → 34; button → 27 and GND; KY-037 D0 → 26.

7. Finger slot: glue the MAX30102 face up at the bottom of a 2-cm slot lined with black foam to block light.
8. Pendant: solder the LiPo to the TP4056 B+/B–, TP4056 OUT+/OUT– to the ESP32-C3 5V and GND, the MPU6050 to 3V3, GND, GPIO 6 (SDA), and GPIO 7 (SCL), the button between GPIO 3 and GND, and the buzzer between GPIO 4 and GND. Fit it in a small box on a breakaway neck cord with the TP4056 USB port reachable.

5. Program the Robot and the Pendant

Set up Telegram for the family

1. In Telegram, open @BotFather, send /newbot, and follow the steps. Copy the bot token it gives.
2. Create a group (e.g., “Bantay kay Lola”), add the family members and the new bot.
3. Send any message in the group, then open https://api.telegram.org/bot<TOKEN>/getUpdates in a browser. The number after "chat":{"id": is the group chat id (it starts with a minus sign).

Upload the programs

1. Install Arduino IDE 2. In Boards Manager, install “esp32 by Espressif Systems” (add https://espressif.github.io/arduino-esp32/package_esp32_index.json in Preferences first).
2. In Library Manager, install: Adafruit MLX90614 Library, SparkFun MAX3010x Pulse and Proximity Sensor Library, Adafruit SSD1306, Adafruit GFX Library, DFRobotDFPlayerMini, UniversalTelegramBot, and ArduinoJson.
3. Open firmware/alaga/alaga.ino. At the top, type the Wi-Fi name and password, the bot token, the group chat id, the older adult's name, and the schedule (hour, minute, task).
4. Board: ESP32 Dev Module. Upload with the wheels lifted. Open Serial Monitor at 115200 and note the Wi-Fi channel it prints. The group should receive “ALAGA is on duty…”.
5. Open firmware/alaga_pendant/alaga_pendant.ino, set WIFI_CHANNEL to that channel, select ESP32C3 Dev Module (USB CDC On Boot: Enabled), and upload.
6. Test: hold the pendant button for 2 s. The robot should say the alert line and the group should get “SOS pressed…”.

If a motor turns backward, swap its two wires at the L298N. **If ALAGA cannot see the tape,** turn each line-sensor trimmer until its LED lights over the tape only; for white tape on a dark floor, change LINE_ON_TAPE to LOW. **If 0x57 is missing in an I²C scan,** buy a MAX30102 board with 3.3 V pull-ups.

Record the voice prompts

Record in a quiet room, save as MP3, and copy to the microSD card (FAT32): prompts in a folder named mp3 with the exact names below, and up to five favorite songs or prayers as 02/001.mp3 to 02/005.mp3.

File	What ALAGA says (Filipino, then English)
mp3/0001.mp3	Magandang araw po! Ako si ALAGA. Good day! I am ALAGA.
mp3/0002.mp3	Gaano po kasakit ang nararamdaman ninyo, mula 0 hanggang 10? Ipihit ang bilog, saka pindutin ang pulang button. How much pain, 0 to 10? Turn the dial, then press the red button.
mp3/0003.mp3	Ilapit po ang noo sa sensor sa aking dibdib. Bring your forehead close to the sensor on my chest.
mp3/0004.mp3	Ilagay po ang hintuturo sa butas at huwag gumalaw. Place your finger in the slot and keep still.
mp3/0005.mp3	Normal po ang inyong mga sukat. Your readings are normal.
mp3/0006.mp3	Sasabihan ko po ang pamilya ninyo. Magpahinga muna po kayo. I will tell your family. Please rest.
mp3/0007.mp3	Tinatawagan ko na po ang pamilya ninyo ngayon. I am alerting your family now.
mp3/0008.mp3	Salamat po! Ingat palagi. Thank you! Take care.
mp3/0009.mp3	Mas lapit pa po nang kaunti. A little closer, please.
mp3/0010.mp3	Hindi ko po nabasa. Subukan ulit. I could not read that. Please try again.
mp3/0011.mp3	Oras na po ng inyong gamot. Pindutin ang pulang button pagkainom. It is time for your medicine. Press the red button after you take it.

mp3/0012.mp3	Paalala po ulit: oras na ng gamot. A reminder again: it is time for your medicine.
mp3/0013.mp3	Salamat po, naitala ko na. Thank you, I have recorded it.
mp3/0014.mp3	Kumusta po kayo? Pindutin ang pulang button kung okay lang kayo. How are you? Press the red button if you are okay.
mp3/0015.mp3	Uminom po tayo ng tubig. Let us drink some water.
mp3/0016.mp3	Paumanhin po, dadaan ako. Excuse me, I am passing through.
mp3/0017.mp3	Narinig ko po kayo. Ipinapaalam ko na sa pamilya ninyo. I heard you. I am alerting your family.
mp3/0018.mp3	Tinawag niyo po ako? Pindutin ang button kung okay kayo. Did you call me? Press the button if you are okay.
mp3/0019.mp3	Ipihit ang bilog para pumili ng kanta o dasal, saka pindutin ang button. Turn the dial to choose a song or prayer, then press the button.
mp3/0020.mp3	Babalik na po ako sa aking puwesto. I am going back to my place.

6. Tape the Home Route and Do the First Run

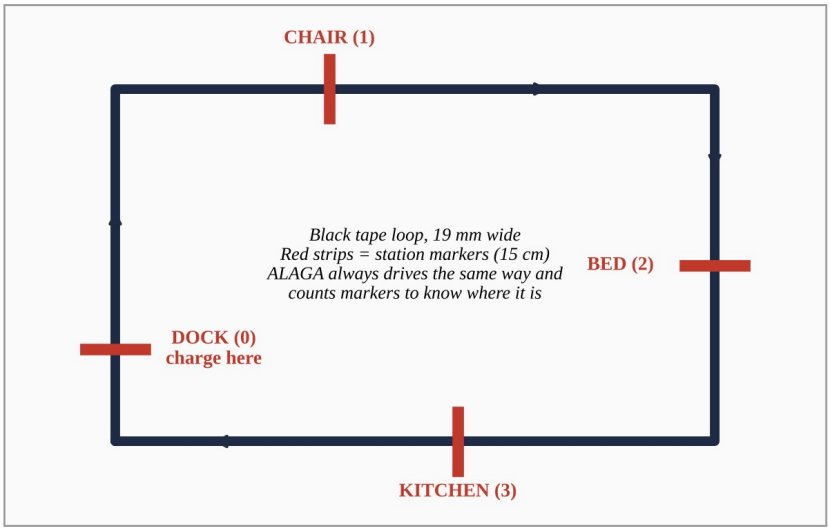


Figure C. Example route: one loop through the dock, chair, bed, and kitchen.

1. Plan one closed loop at least 50 cm from walls and furniture legs, with gentle corners (radius 30 cm or more). Lay 19-mm black electrical tape on a clean, light floor; press it flat so no one trips.
2. Mark each station with a 15-cm strip across the line, in this order around the loop: dock (0), chair (1), bed (2), kitchen (3). ALAGA always drives the same way around the loop.
3. Put ALAGA on the line just past the dock marker, facing the chair. Switch on. Hold the red button for 5 s to tell it that it is at the dock.
4. Send /check in the Telegram group. ALAGA inflates (30 s), drives to the chair, asks “Kumusta po kayo?”, and waits for the button. Press it; the group gets the answer, and ALAGA goes around the loop to the dock and deflates.
5. Put a box on the line: ALAGA must stop about 25 cm before it and say “Excuse me”. Push the bumper: it must stop and back up a little.
6. Send /vitals and do a full health check; then /status to see the last results.
7. Wear the pendant and drop onto a gym mat (researchers only). After the 10-s beep, the group must get a FALL alert. Repeat and press the pendant button during the beep: no alert should be sent.
8. Charge the robot battery every night and the pendant every 2–3 days.

7. Calibrate and Test

1. Bench test: heat water with a black-painted can and a probe thermometer; take 10 ALAGA readings at 35.0, 36.0, 37.0, 37.5, 38.0, and 40.0 °C.
2. Calibration set (10 volunteers, after IRB approval): three ALAGA and three reference readings each. In Excel, =SLOPE(reference, raw) gives TEMP_A and =INTERCEPT(reference, raw) gives TEMP_B (set TEMP_A = 1 and TEMP_B = 0 while collecting raw readings). Average (reference − ALAGA) for PR_OFFSET and SPO2_OFFSET.

3. Fall thresholds: with the first 10 simulated falls and 10 daily activities (sit, lie on a bed, bend, walk, climb a step), check the peak and low g values in the pendant's Serial Monitor and adjust FREE_FALL_G and IMPACT_G if needed. Then keep them fixed for the real tests.
4. Run the trials in the paper (Table 7): navigation trips, validation health checks, falls and activities, 5 days of reminders, and the soft-versus-rigid usability test (rigid shell: a coroplast cylinder with the same chest box and face).

8. Troubleshooting and Care

Problem	Likely cause	Fix
Nothing turns on	Fuse blown, switch off, battery flat	Check fuse; charge battery
Robot spins or leaves the line	Motor wires swapped; speed too high	Swap wires; lower SPEED or KP
Stops at the wrong station	A marker was missed or doubled	Make markers 15 cm; fix tape gaps
No Telegram messages	Wrong token or chat id; Wi-Fi down	Check settings; Serial Monitor shows SENT
Pendant alerts not received	Wrong Wi-Fi channel in the pendant	Use the channel the robot prints
Too many false falls	Thresholds too low	Raise IMPACT_G a little; re-test
Temperature reads too low	Too far, sweat, not calibrated	3–5 cm; wipe forehead; calibrate
Body goes soft	Big leak at a seam or the hem	Seal seam; re-staple the hem

- Wipe the forehead window and finger slot with 70% alcohol after each use.
- Keep the tape route clean and flat; replace torn tape.
- ALAGA helps the family watch over Lola or Lolo; it never replaces a visit, a call, or a doctor.